

Conglomerate Mergers: Effects on Growth and Competition

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Abstract

Your abstract goes here. It should be a brief summary of your article.

1 Introduction

[2]

2 Background

Compound annual growth rate volatility tax, variance drain [1], [5] [3] Merge as a hedge against larger draw downs

$$g = \mu - \frac{\sigma^2}{2} \quad (1)$$

3 Model

3.1 Random Multiplicative Growth

3.1.1 Growth Process

To investigate conglomerate formation we set up an Agent Based Model with M markets that are uncorrelated between each other, each containing an amount of N firms. Firms face random multiplicative growth that is unique for their market, drawn from a normal distribution with positive mean $\mathcal{N}(\mu_m, \sigma_m)$. In this framework each firms state is multiplicatively linked to its previous one. This can be interpreted as firms setting a certain price and obtaining a random demand for it.

$$s_{it+1} = s_{it}\delta_t \quad \text{with} \quad \delta_t \sim \mathcal{N}(\mu_m, \sigma_m) \quad (2)$$

We model markets such that mean and the standard deviation of the distribution over their growth increments are positively correlated, meaning that a higher mean μ_m comes with a higher standard deviation σ_m . As explained in section 2, in multiplicative processes one is not just maximizing for the ensemble mean, but for the growth rate. Hence, a higher scale σ_m decreases the growth rate, through the volatility tax.

Each firm thus realizes a gain of $\Delta_{it} = s_{it}\delta_t - s_{it-1}$. For any firm, not part of a conglomerate this will also be their period profit $\Pi_{it} = \Delta_{it}$. Firms in conglomerates pool an amount $\alpha\Delta_{it}$, which flows into the conglomerate's period pool $\Omega_{ct} = \sum_{i \in c} \alpha\Delta_{it}$. However, as conglomerates grow in size, maintaining them becomes ever more difficult as transaction costs increase. We model these as a logistic function of the form

$$\Phi(O_c, M) = \frac{1}{1 + \exp^{-|O_c| + 0.4M}} \quad (3)$$

This factor scales the cost according to conglomerate size $\sum_{i \in c} s_{it}$ such that the pool size is given by

$$\Omega_{ct} = \sum_{i \in c} \alpha \Delta_{it} - \Phi(O_c, M) \sum_{i \in c} s_{it} \quad (4)$$

Notice that the factor 0.4 in logistic function scales, the costs so that a conglomerate operating in 40 % of all markets would lose 50 % of their market value in transaction cost every round. Lastly, this pool will be shared equally such that a members profit is determined by $\Pi_{it} = (1 - \alpha)\Delta_{it} + \frac{\Omega_{ct}}{|O_{c_i}|}$. Consequently, any firms next state is given by $s_{it+1} = s_{it} + \Pi_{it}$.

3.1.2 Merging Process

At the beginning of every time step t each firm $i \in M \times N$ gets a chance to choose another with probability p . If successful, it will select a random other firm j , among the remaining $(M \times N) - 1$ firms. We allow conglomerates C to only operate one firm per market, meaning it can have a maximum of M members, with the set markets of a conglomerate defined as $O_c \subseteq M$. Hence, conglomeration is only *feasible* if firm j and i :

1. don't operate in the same market ($m_i \neq m_j$)
2. are not part of a conglomerate that operates in a market in which the other firm, or the conglomerate which it is part of operates ($O_{c_i} \cap O_{c_j} = \emptyset$)

We assume that if both firms are in conglomerates that these will merge as well. After confirming *feasibility*, merging firms determine desirability, meaning that they would both benefit from joining forces. They do so by comparing the realized pool Ω size over the last l periods, against a "synthetic" pool $\hat{\Omega}$ size if they had merged. If $\hat{\Omega} > \Omega$ the merger will be implemented.

3.1.3 Splitting Process

After an initial l periods each firm that is part of a conglomerate evaluates its performance within it by comparing its growth rate $g_{i(t-l):t}$ by being part

of the conglomerate against the potential rate $\hat{g}_{i(t-l):t}$ it had realized, had it remained outside. Upon being worse off, a firm exits the conglomerate and continues to operate alone.

Algorithm 1: Model flow with random growth.

```

for  $t$  in  $T$  do
  foreach  $i \in N \times M$  do
     $\lfloor$  Merge with  $j$  if  $(m_i \neq m_j)$  and  $(O_{c_i} \cap O_{c_j} = \emptyset)$  and  $\hat{\Omega} > \Omega$ 
  foreach  $c \in C$  do
     $\Omega_{ct} = \sum_{i \in c} \alpha \Delta_{it} - \Phi(O_c, M) \sum_{i \in c} s_{it}$ 
    foreach  $i \in c$  do
       $\lfloor$   $\Pi_{it} = (1 - \alpha) \Delta_{it} + \frac{\Omega_{ct}}{|O_c|}$ 
    foreach  $i \notin c$  for all  $c \in C$  do
       $\lfloor$   $\Pi_{it} = \Delta_{it}$ 
    foreach  $i \in c$  for all  $c \in C$  do
       $\lfloor$  Exit  $c$  if  $\hat{g}_{i(t-l):t} > g_{i(t-l):t}$ 

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3.2 Endogenous Pricing

Going beyond random growth, we enable our firms to optimize revenues by setting their period prices p_{it} for which they receive the following demand

$$D_{it}(\mathbf{P}_t, \theta_{it}) = 1 - (1 - \theta_t)p_{it} + 0.5 \max_{j:j \neq i} \mathbf{P}_t \quad (5)$$

where θ_{it} is the firms current market share, which are the relative cumulative profits within a market

$$\theta_{it} = \frac{\sum_{\tau=1}^t \Pi_{i\tau}}{\sum_{\tau=1}^t \sum_{j:j \neq i} \Pi_{j\tau}} \quad (6)$$

Consequently, period revenues take the form of $D_{it}(\mathbf{P}_t, \theta_{it})p_{it}$. We further assume stochastic costs γ that are drawn from a log-normal distribution $\mathcal{L}(0, 1)$ that is scaled to match the mean growth rate of revenues per period. Hence, a firm's pre-pooling profit is given by $\Delta_{it} = D_{it}(\mathbf{P}_t, \theta_{it})p_{it} - \gamma$. If the firm is not in a conglomerate, there is no pooling, meaning $\Pi_{it} = \Delta_{it}$. In case it is part of a conglomerate, it will continue as outlined in section 3.1.

To learn the optimal pricing policy π , we trained a reinforcement learning agent using the Proximal Policy Optimization algorithm [4]. We trained the agents in a separate environment being tasked to only optimize revenues

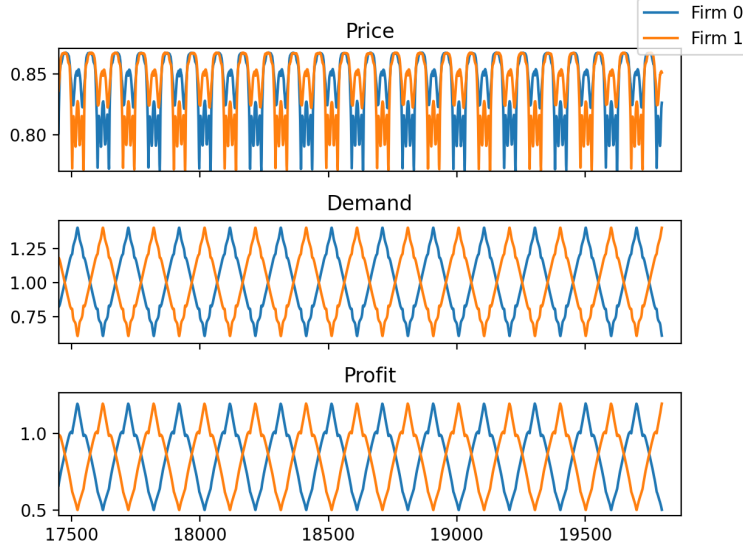


Figure 1: Testing a trained policy with countervailing market shares.

$D_{it}(\mathbf{P}_t, \theta_{it})p_{it}$. Its information set contains its price and market share in the previous round, as well as the maximum price of its competitors in the previous two rounds, $\{p_{it-1}, \theta_{it-1}, \max_{j:j \neq i} \mathbf{P}_{jt-1}, \max_{j:j \neq i} \mathbf{P}_{jt-2}\}$. To expose the agents to possible market share constellations, we exogenously set the θ_{it} . The resulting policy can be seen in figure 1.

The resulting model flow can be seen in algorithm 2.

Algorithm 2: Model flow with endogenous pricing.

```

for  $t$  in  $T$  do
  foreach  $i \in N \times M$  do
     $\lfloor$  Merge with  $j$  if  $(m_i \neq m_j)$  and  $(O_{c_i} \cap O_{c_j} = \emptyset)$  and  $\hat{\Omega} > \Omega$ 
  foreach  $i \in N \times M$  do
     $\lfloor$   $p_{it} = \pi(p_{it-1}, \theta_{it-1}, \max_{j:j \neq i} \mathbf{P}_{jt-1}, \max_{j:j \neq i} \mathbf{P}_{jt-2})$ 
   $\Delta_t = D(\mathbf{P}_t, \theta_t) \mathbf{P}_t - \gamma$ 
  foreach  $c \in C$  do
     $\lfloor$   $\Omega_{ct} = \sum_{i \in c} \alpha \Delta_{it} - \Phi(O_c, M) \sum_{i \in c} s_{it}$ 
    foreach  $i \in c$  do
       $\lfloor$   $\Pi_{it} = (1 - \alpha) \Delta_{it} + \frac{\Omega_{ct}}{|O_c|}$ 
  foreach  $i \notin c$  for all  $c \in C$  do
     $\lfloor$   $\Pi_{it} = \Delta_{it}$ 
  foreach  $i \in M \times N$  do
     $\lfloor$   $\theta_{it} = \frac{\sum_{\tau=1}^t \Pi_{i\tau}}{\sum_{\tau=1}^t \sum_{j:j \neq i} \Pi_{j\tau}}$ 
  foreach  $i \in c$  for all  $c \in C$  do
     $\lfloor$  Exit  $c$  if  $\hat{g}_{i(t-l):t} > g_{i(t-l):t}$ 

```

4 Results

4.1 Random Multiplicative Growth

We implemented the models outlined in the previous section in python with a parameter specification that can be found in Table 1. However, note that we conducted extensive robustness testing, none of which altered the central findings of this section.

Parameter	Value
M	100
N	100
T	1000
l	50
p	0.05
α	$[0, 0.18]$

Table 1: Model Parameters and their values

To answer the questions outlined in section 2, we analyze certain outputs

of our model, which can be seen in figure 2. Firstly, panels (a)–(c) show a stable tendency to form conglomerates, which after initial fluctuations, partially due the review period l , settles to a stable level. Both the number of conglomerates, and their size converge to a regular pattern. In addition, when looking at panels (d) and (e), which are snapshots taken after the model reached its ultimate iteration, we see that that larger conglomerates are made up of firms that on average occupy higher ranks and have higher market share in their respective markets. Note that a firm’s rank is determined ordering each firm in a market by market share. Panel (f) in turn fits a polynomial of degree two to the relation between conglomerate size and average market share $mean_share = \beta_0 + \beta_1 size + \beta_2 size^2$, the result of which indicates an increasing importance of being in a larger conglomerate, given the steady increase in β_1 . These findings suggest that our proposed mechanism behind conglomerate formation explains a part of this phenomenon. However, as a conglomerate’s size increases we see the transaction costs biting into profits, capping its growth as β_2 becomes ever more negative. Consequently, this sheds light on our question 3, by setting an upper bound on conglomerate size.

Thirdly, the last row of figure 2 provides insights into the competitive landscape. Panel (g) showcases how firms move between market share percentiles. While almost every firm was above the 10 % level, the top spot is occupied by a few firms for the majority of the time. This suggests that there is some mobility but tendency to have strong agglomeration effects. Panel (h) yields further evidence to the point as we can see the firm with the highest market share averaging roughly 20 % throughout the model’s runtime. Lastly, we see that the Gini coefficient calculated for each market steadily increases over time, suggesting a rise in market share inequality.

5 Discussion

6 Conclusion

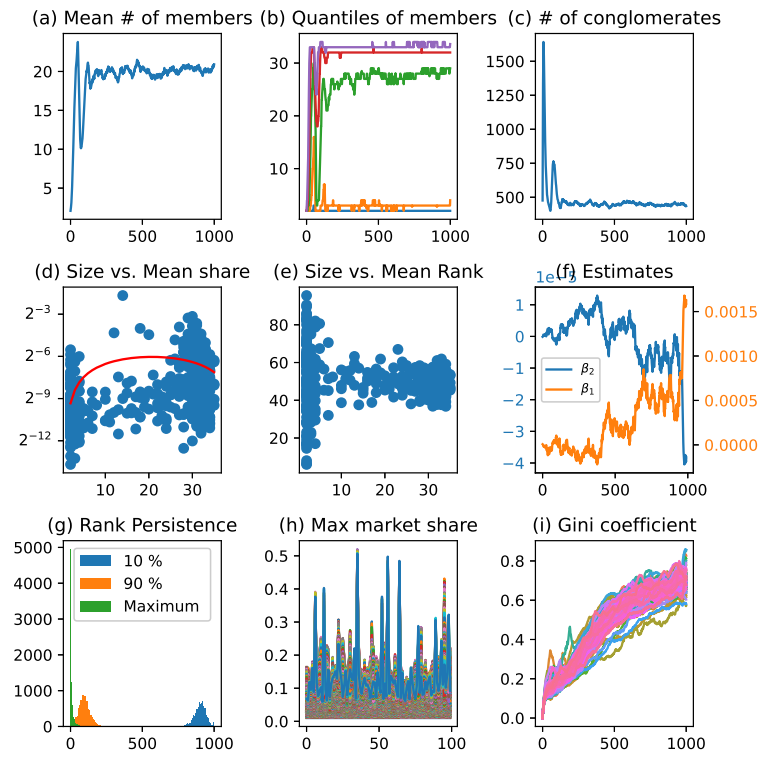


Figure 2: Overview of relevant variables ($\alpha = 0.2$).

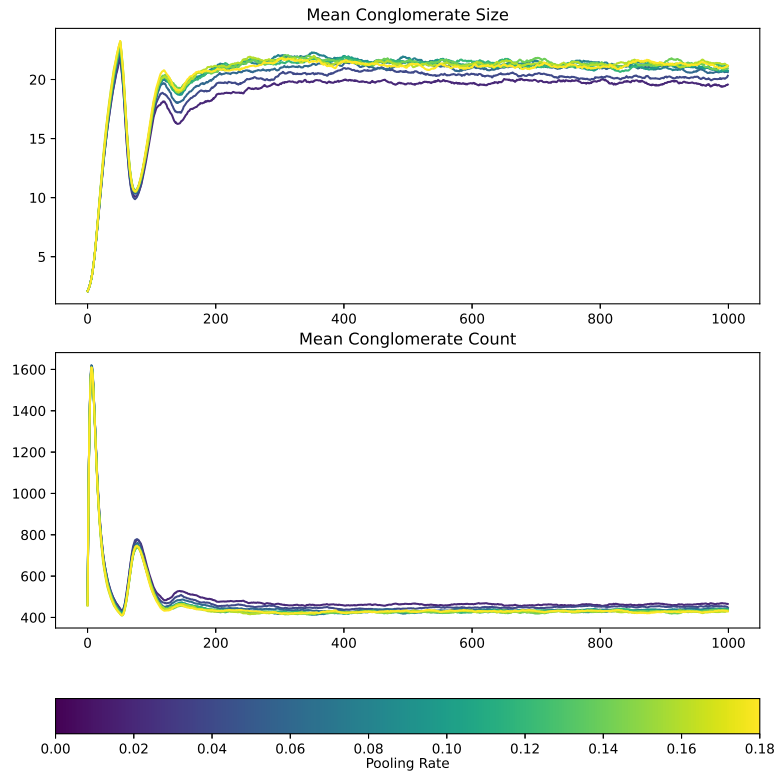


Figure 3: Conglomerate sizes and number of conglomerates $\alpha \in [2, 0.18]$.

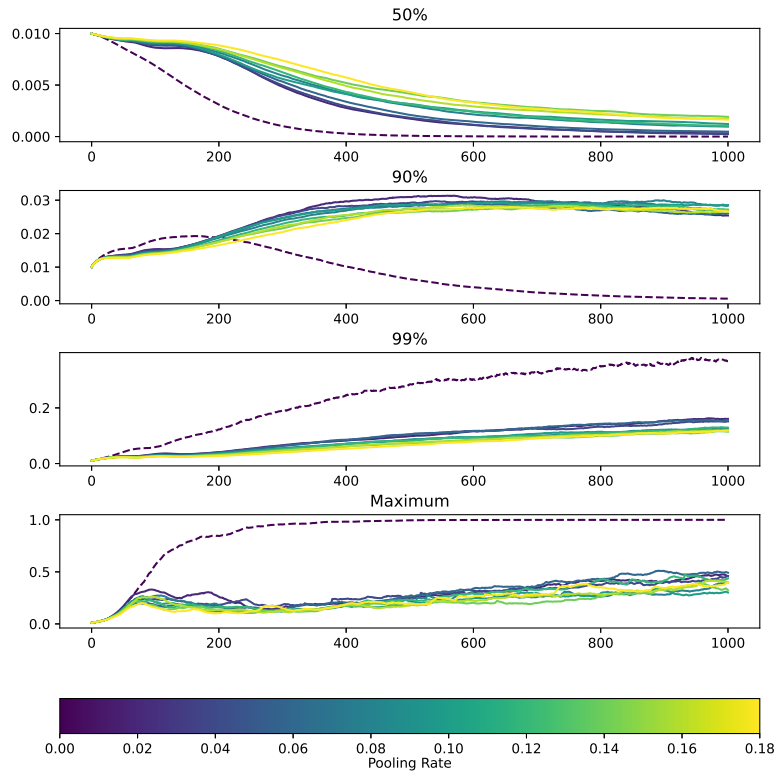


Figure 4: Market share of a given quantile $\alpha \in [0, 0.18]$.

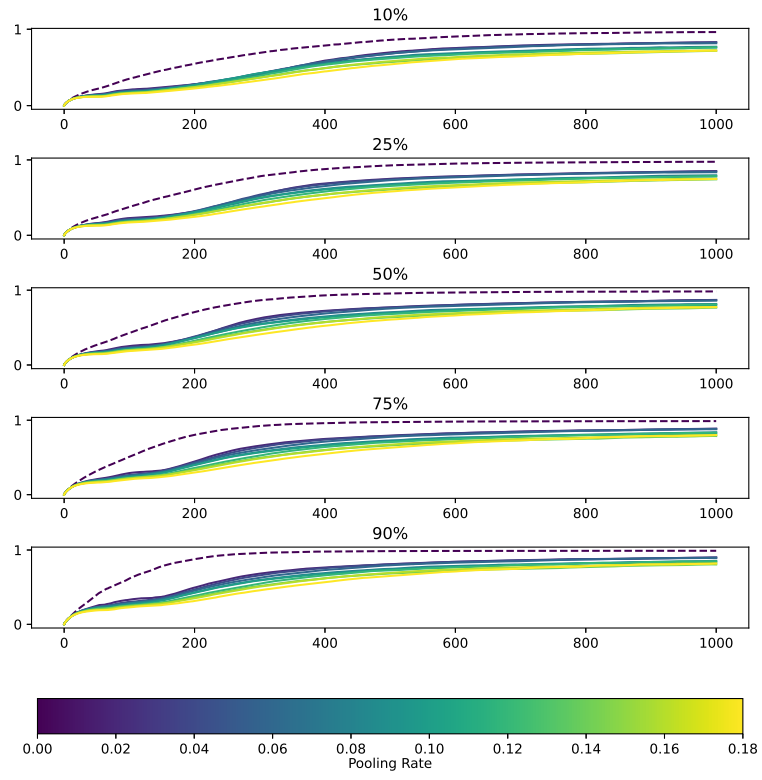


Figure 5: Quantiles of Gini coefficients $\alpha \in [0, 0.18]$.

References

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